

FASTEST FINGER FIRST SYSTEM

Using Raspberry Pi

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Fastest finger first systems are used in school, college and TV contests. The team which presses their push-button switch first is entitled to answer the query. When two or more teams press their push-buttons within a very small time gap, it becomes very difficult to identify the team that pressed the button first. In such cases, the decision can be biased due to human intervention.

The system presented here takes care of the aforesaid problem. This system can be used for a maximum of four teams in its present form. It is built around a Raspberry Pi board and a solid-state relay module.

Circuit and working

The Raspberry Pi runs a standard Raspbian Linux distribution with Wi-Fi dongle, installed GPIO library and software written in Python. Most of the coding has been done on the Raspberry Pi. Indicator bulbs are provided on contestants' desks to know who pressed the push-buttons.

The HDMI output is connected to an LCD monitor to know the sequence of buttons pressed by the teams if two or more teams press the buttons almost simultaneously. The circuit can also be controlled via an SSH connection using VX ConnectBot software on an Android smartphone.

You may refer previous EFY issues for installation of standard Raspbian Linux distribution. Installation guide for Wi-Fi dongle on Raspberry Pi varies from dongle to dongle, so you may check it on Google.

Raspberry Pi GPIO library can be installed using following commands on the terminal window:

```
$ sudo apt-get install python3-rpi.gpio
```

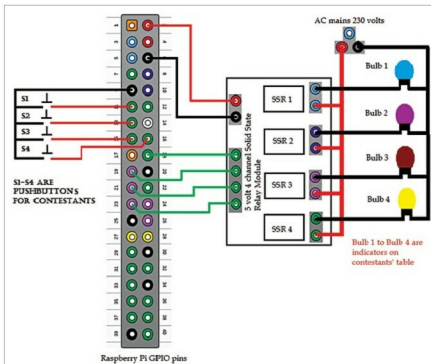


Fig. 1: Circuit diagram of the fastest finger first system using Raspberry Pi

```
OR
$ sudo apt-get install python3-rpi.gpio
```

To access the above libraries add following lines to the Python script (rasquiz.py):

```
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BOARD)
termcolor module is used for
simple formatting of text to colour,
make bold, underline, highlight, etc.
Install termcolor using command
given below:
```

```
$ wget http://pypi.python.org/packages/
source/t/termcolor/termcolor-1.1.0.tar.
```

```
gz
$ tar -xvzf termcolor-1.1.0.tar.gz
$ cd termcolor-1.1.0
```

```
$ sudo python setup.py install
```

Please refer the link below for more detail.

<https://pypi.python.org/pypi/termcolor>

Download VX ConnectBot software from Play Store on Android smartphone. Start tethering or hotspot on smartphone and connect Raspberry Pi to it over a Wi-Fi network. After that start VX ConnectBot software and login to Raspberry Pi over SSH connection using `pi@ <ip_address>` and default password (in this case `raspberrypi`). A terminal window will open on your Android phone from where you can run the Python script.

The input push-button switches (S1 through S4) are wired such that when pressed, its one end gets connected to GND (pin 9). The input pins are pulled up (internally) to 3.3V by the optional argument:

```
pull_up_down=GPIO.PUD_UP in GPIO.setup
```